

What is the difference between a trajectory and an equilibrium point in a phase portrait of an SDE?

Answer:

In the context of a phase portrait of a system of differential equations (SDE), the terms "trajectory" and "equilibrium point" have specific meanings:

1. Trajectory:

- A trajectory represents a path traced by the solution of a system over time in the phase plane. It shows how the state of the system evolves starting from a particular initial condition.
- Trajectories are curves, and they change as time progresses. The direction of the trajectory is often indicated by arrows, demonstrating how the state moves through the phase plane.

2. Equilibrium Point:

- An equilibrium point is a point in the phase plane where the system remains constant over time. This occurs when the derivative of the system at that point is zero, indicating no change in state.
- Unlike trajectories, equilibrium points do not trace out paths; they are static. The most common equilibrium point, often referred to in examples, is the origin where the system's general solution reduces to a constant.

In summary, both a trajectory and an equilibrium point represent a particular solution of an SDE. A trajectory is dynamic and shows the system's evolution over time, while an equilibrium point is static and represents a state where the system remains unchanged.